

Chapter 8 Quiz Review

1. $\int 5x^4 \ln x dx$

2. $\int \frac{4x+3}{x^2+25} dx$

3. $\int x^2 \sin x dx$

4. $\int x \sin x^2 dx$

5. $\int \frac{x+10}{x^2-x-2} dx$

6. $\int \frac{9}{x^2-8x+20} dx$

$$7. \int 3xe^{3x} dx$$

$$8. \int \frac{18x-11}{6x^2-13x-5} dx$$

$$9. \int \frac{7x^2+12}{x^3+x^2-6x} dx$$

$$10. \int_1^{e^2} \ln x dx$$

$$11. \int_0^1 \frac{2x-3}{\sqrt{4-x^2}} dx$$

$$12. \int_4^5 \frac{-dx}{x^2-4x+3}$$

Answers

1. This is integration by parts and ILATE says $u = \ln x$, $dv = 5x^4 dx$. So $du = \frac{1}{x} dx$, $v = x^5$ and $\int 5x^4 \ln x dx = x^5 \ln x - \int x^5 \frac{1}{x} dx$. Simplifying the integrand gives us $x^5 \ln x - \int x^4 dx$ which can be integrated as is, so $x^5 \ln x - \frac{1}{5} x^5 + C$.
2. The denominator can't be factored, so this isn't partial fractions. Separating the numerator and pulling out the constant gives us $4 \int \frac{x}{x^2 + 25} dx + 3 \int \frac{1}{x^2 + 25} dx$. The first integral looks like $\ln$, where $u = x^2 + 25$, $du = 2x dx$. So $\frac{4}{2} \int \frac{du}{u} = 2 \ln|x^2 + 25| + C$. The second integral looks like $\arctan$, where $u = x$, $du = dx$, $a = 5$. So, $3 \int \frac{du}{a^2 + u^2} = (3) \frac{1}{5} \arctan\left(\frac{x}{5}\right) + C$ and $2 \ln|x^2 + 25| + \frac{3}{5} \arctan\left(\frac{x}{5}\right) + C$.
3. This is integration by parts and ILATE says $u = x^2$, $dv = \sin x dx$. So $du = 2x dx$, $v = -\cos x$ and $\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx$. This new integral requires another integration by parts and ILATE says $u = 2x$, $dv = \cos x dx$. So $du = 2 dx$, $v = \sin x$ and $\int 2x \cos x dx = 2x \sin x - \int 2 \sin x dx$. This integral can be integrated as is, so $\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$.
4. This is a u-substitution with $u = x^2$, $du = 2x dx$. So $\frac{1}{2} \int \sin u du = -\frac{1}{2} \cos x^2 + C$.
5. Since the denominator can be factored, this could be partial fractions. $\frac{x+10}{x^2 - x - 2} = \frac{A}{x-2} + \frac{B}{x+1}$ and $x+10 = A(x+1) + B(x-2)$. Plugging in $x = -1$, $x = 2$ gives $9 = -3B$, $12 = 3A$ and $B = -3$, $A = 4$. So, $\int \frac{x+10}{x^2 - x - 2} dx = 4 \int \frac{1}{x-2} dx + -3 \int \frac{1}{x+1} dx$. These both are $\ln$ so $4 \ln|x-2| - 3 \ln|x+1| + C$.

6. Since the denominator cannot be factored, we should look to complete the square. So, we should take half the middle coefficient, square it $\left(\frac{-8}{2}\right)^2 = (-4)^2 = 16$ and then add and subtract this 16 in the denominator. This gives us $\frac{9}{x^2 - 8x + 16 + 20 - 16}$. We can then rewrite the first 3 terms as a perfect square and then simplify the last 2 terms, giving us $\int \frac{9}{(x-4)^2 + 4} dx$. This is then set up to be arctan where $u = x - 4, du = dx, a = 2$. So, $\int \frac{9}{(x-4)^2 + 4} dx = \frac{9}{2} \arctan\left(\frac{x-4}{2}\right) + C$.
7. This is integration by parts and ILATE says $u = 3x, dv = e^{3x} dx$. So $du = 3dx$ and finding v will require a u-substitution, where $u = 3x, v = \frac{1}{3} \int e^u du$ or $v = \frac{1}{3} e^{3x}$. So, $\int 3xe^{3x} dx = xe^{3x} - \int e^{3x} dx$. This new integral requires a u-sub with $u = 3x, du = 3dx$. So $\int 3xe^{3x} dx = xe^{3x} - \frac{1}{3} \int e^u du$ or $xe^{3x} - \frac{1}{3} e^{3x} + C$.
8. Since the denominator can be factored, this could be partial fractions. $\frac{18x-11}{6x^2-13x-5} = \frac{A}{2x-5} + \frac{B}{3x+1}$ and $18x-11 = A(3x+1) + B(2x-5)$. Plugging in $x = -\frac{1}{3}, x = \frac{5}{2}$ gives $-17 = -\frac{17}{3}B, 34 = \frac{17}{2}A$ and $B = 3, A = 4$. So, $\int \frac{2x+9}{x^2+4x-21} dx = 4 \int \frac{1}{2x-5} dx + 3 \int \frac{1}{3x+1} dx$. These both are ln and both require a u-sub where $u = 2x-5, du = 2dx$ and $u = 3x+1, du = 3dx$. So, $\frac{4}{2} \int \frac{du}{u} + \frac{3}{3} \int \frac{du}{u}$ gives us $2 \ln|2x-5| + \ln|3x+1| + C$.
9. Since the denominator can be factored, this could be partial fractions. $\frac{7x^2+12}{x^3+x^2-6x} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-2}$ and $7x^2+12 = A(x+3)(x-2) + Bx(x-2) + Cx(x+3)$. Plugging in $x = 0, x = -3, x = 2$ gives $12 = -6A, 75 = 15B, 40 = 10C$ and $A = -2, B = 5, C = 4$. So, $\int \frac{7x^2+12}{x^3+x^2-6x} dx = -2 \int \frac{1}{x} dx + 5 \int \frac{1}{x+3} dx + 4 \int \frac{1}{x-2} dx$. These are all ln and don't require a u-sub where any constants are needed. So, $\int \frac{7x^2+12}{x^3+x^2-6x} dx = -2 \ln|x| + 5 \ln|x+3| + 4 \ln|x-2| + C$.

10. Since this can't be integrated as is, this is integration by parts. ILATE says $u = \ln x$, $dv = dx$. So,

$$du = \frac{1}{x} dx, v = x \text{ and } \int_1^{e^2} \ln x dx = x \ln x - \int_1^{e^2} x \left(\frac{1}{x} \right) dx, \text{ which simplifies to } x \ln x - \int_1^{e^2} dx = x \ln x - x \Big|_1^{e^2}.$$

evaluating at the bounds gives $(e^2 \ln e^2 - e^2) - (1 \ln 1 - 1) = 2e^2 - e^2 + 1 = e^2 + 1$.

11. Separating the numerator and pulling out the constant gives us $2 \int \frac{x}{\sqrt{4-x^2}} dx - 3 \int \frac{1}{\sqrt{4-x^2}} dx$. The first

integral looks like a u-substitution, where $u = 4 - x^2$, $du = -2x dx$. So $-\frac{2}{2} \int \frac{du}{\sqrt{u}} = -2\sqrt{4-x^2}$. The

second integral looks like arcsin, where $u = x$, $du = dx$, $a = 2$. So, $-3 \int \frac{du}{\sqrt{a^2 - u^2}} = -2 \arcsin\left(\frac{x}{2}\right)$ and

$$-2\sqrt{4-x^2} - 3\arcsin\left(\frac{x}{2}\right) \Big|_0^1 = \left(-2\sqrt{3} - 3\arcsin\left(\frac{\pi}{6}\right)\right) - (-4 - 3\arcsin 0), \text{ or } -2\sqrt{3} - \frac{\pi}{2} + 4.$$

12. Since the denominator can be factored, this could be partial fractions. $\frac{-1}{x^2 - 4x + 3} = \frac{A}{x-3} + \frac{B}{x-1}$ and

$$-1 = A(x-1) + B(x-3). \text{ Plugging in } x=1, x=3 \text{ gives } -1 = -2B, -1 = 2A \text{ and } B = \frac{1}{2}, A = -\frac{1}{2}.$$

So, $\int_4^5 \frac{-1}{x^2 - 4x + 3} dx = -\frac{1}{2} \int_4^5 \frac{1}{x-3} dx + \frac{1}{2} \int_4^5 \frac{1}{x-1} dx$. These both are ln with no u-sub so

$$-\frac{1}{2} \ln|x-3| + \frac{1}{2} \ln|x-1| \Big|_4^5 = \left(-\frac{1}{2} \ln 2 + \frac{1}{2} \ln 4\right) - \left(-\frac{1}{2} \ln 1 + \frac{1}{2} \ln 3\right) \text{ which simplifies to}$$

$$-\frac{1}{2} \ln 2 + \frac{1}{2} \ln 4 - \frac{1}{2} \ln 3 = \frac{1}{2} (\ln 4 - \ln 2 - \ln 3) = \frac{1}{2} \ln\left(\frac{4}{6}\right) = \frac{1}{2} \ln\left(\frac{2}{3}\right) = \ln \sqrt{\frac{2}{3}}.$$